

MAHENDRA PRATAP SINGH

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PROFESSIONAL SUMMARY

- Over **20 years of experience** in **Test Automation**, **Test Management**, and **Quality Assurance** across diverse industries, leveraging **Agile**, **SAFe**, and **Waterfall** methodologies.
 - **10 years of test management experience**, leading **cross-functional teams** of 20+ professionals across multiple geographies, ensuring seamless delivery of complex projects.
 - **16+ years** of hands-on experience in **Test Automation Architecture**, driving the adoption of **shift-left testing** principles by designing scalable frameworks for **web**, **mobile**, **API testing**.
 - Experience developing custom **automation utilities for validating HIPAA Electronic Data Interchange (EDI) files** for healthcare industry to exchange administrative and financial data.
 - Expertise in **Test Strategy Development**, **Risk-Based Testing**, **Defect Management**, and **Quality Governance**, ensuring **alignment with business objectives and industry standards**.
 - Proven track record in **mentoring and leading** QA engineers and test automation specialists, fostering a culture of collaboration, innovation, and quality excellence.
 - Industry expertise includes **Banking** (Digital Applications, Payments & Cards, Retail Tools), **Health Insurance** (Claims and Payment), **Land Information** and **Telecom**.
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CORE COMPETENCIES

- **Test Strategy & Planning.**
 - **Automation Framework Design (Web, Mobile, API).**
 - **CI/CD Integration and Cloud Technologies.**
 - **Test Management & Agile Delivery.**
 - **Client & Stakeholder Management.**
 - **Team Leadership, Mentorship & Development.**
 - **Quality Assurance & Continuous Improvement.**
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TECHNICAL SKILLS

Test Automation & Tools:

- **Web Application Testing:** Playwright, Cypress, React Testing Library, Selenium WebDriver, Browser Use, Selenium Basic (Excel Macro Plugin).
- **API Testing:** REST Assured, Postman, Spring Boot, OpenAPI, Swagger, Supertest, SoapUI, Insomnia, Karate.
- **Mobile Testing:** XCUITest (iOS), Espresso (Android), Appium, Experitest SeeTest Automation, Appium Studio.

- **Automation Frameworks:** TestNG, JUnit, Jest, Mocha, Cucumber, Concordion, PyTest.
- **Desktop Application Testing:** UFT 11.0, QuickTest Professional (QTP) 9+, WinRunner 8.2.
- **Test Automation Architecture:** Keyword driven, Data driven, Page Object Model, Behavior Driven and Hybrid.
- **CI/CD Tools:** Jenkins, GitHub Actions, Azure DevOps.
- **Version Control:** Git, GitHub, GitLab, Bitbucket, Stash, WinCVS, TortoiseSVN.
- **Defect Management:** ALM/Quality Center, Change Synergy.
- **Build Tools:** Maven, Gradle, NodeJS.

Programming Languages:

- TypeScript, JavaScript, Java, Swift, Kotlin, Python, VB.NET, C#, C/C++, VBScript, Test Script Language(TSL).

Agile & Management Tools:

- Waterfall, Agile, Scrum, Kanban, SAFe Methodology, Jira, Confluence, Miro, InVision. Sharepoint

Cloud & Containerization:

- AWS, Azure, Firebase, Docker, Kafka.

Databases:

- Informix, AWS DynamoDB, Oracle, SQL.

Web & Mobile Development:

- React, Spring Framework, UIKit, SwiftUI, Combine Framework.

Content Management Systems (CMS);

- Vignette CMS, Adobe Analytics, SharePoint.

WORK EXPERIENCE

Senior Test Automation Engineer (Contract)

Reserve Bank of New Zealand (RBNZ)

April 2025 – Present

- Leading the design and development of scalable **API test automation frameworks**, seamlessly integrating them into **Azure DevOps CI/CD pipelines** to support continuous testing and early defect detection.
- Defining and implementing a comprehensive **test data management strategy**, including data **provisioning and masking**, to ensure consistent, reusable, and secure test data across multiple test environments.

Senior Test Automation Engineer (Contract)

Land Information New Zealand (LINZ)

November 2022 – March 2025

- **Developing and testing** front-end applications using **React, React Testing Library**, and **TypeScript** enhancing user experience and functionality.

- **Implementing and testing** back-end solutions using **Spring boot** and **Kotlin**, ensuring seamless system integrations and scalability.
- Leveraged **Swagger and OpenAPI** Specifications for API design & documentation.
- **Conducted end-to-end testing** leveraging **Playwright** to ensure comprehensive application quality and minimize defects in production.
- **Managed CI/CD pipelines** and version control with **Git, GitHub, GitHub Actions, Docker, and AWS**, driving faster delivery and reliable delivery cycles.
- **Applied SAFe Agile methodologies** using Jira, Confluence, Miro, and Figma, fostering collaboration and alignment across teams.

Senior Test Automation Engineer (Contract)

New Zealand Customs Service

July 2022 - November 2022

- Led test automation activities, including designing, developing, and maintaining **automation frameworks and scripts**.
 - **Web Applications** using **Selenium, Cucumber, Cypress**.
 - **API Testing** using **REST Assured**.
 - Building **Postman collections** to use in manual testing.
- Implemented **Java-based tool for bulk API data creation**, achieving a **60% reduction in effort**.
- **Integrated with CI/CD pipelines** using **Jenkins, Git**.
- **Applied SAFe Agile methodologies** using Jira, Confluence, fostering collaboration and alignment across teams.

Senior Test Automation Engineer (Contract)

Bank of New Zealand Limited

December 2021 - July 2022

- Led test automation activities, including designing, developing, and maintaining **automation frameworks and scripts for API testing** using **TypeScript, Jest, Supertest, AWS SDK, AWS DynamoDB, Kafka**.
- Utilized **Swagger and OpenAPI** Specifications for API design & documentation
- Building **Postman collections** to use in manual testing.
- **Integrated with CI/CD pipelines** using **Jenkins, Git**.
- **Applied SAFe Agile methodologies** using Jira, Confluence.

Senior Test Automation Engineer (Contract)

Land Information New Zealand (LINZ)

April 2021 - December 2021

- Performed component testing of **React** front-end applications using **React Testing Library**.
- Developed and executed **unit and integration tests for backend APIs** with **Spring Boot and Kotlin**.
- Leveraged **Swagger and OpenAPI** Specifications for API design & documentation.

- **Conducted end-to-end testing** using **Playwright** and Cypress.
- **Managed CI/CD pipelines** and version control with **Git, GitHub, Jenkins**.
- **Applied SAFe Agile methodologies** using Jira, Confluence, Miro, and Figma, fostering collaboration and alignment across teams.

Senior Test Automation Engineer (Contract)

Bank of New Zealand Limited

August 2019 - April 2021

- Led test automation activities, including designing, developing, and maintaining **automation frameworks and scripts**.
 - **Mobile Application Test Automation** using **XCUITest, Espresso**.
 - **Web Applications** using **Selenium, Cypress**.
 - **API Testing** using **REST Assured, Supertest, Jest**.
 - Building **Postman collections** to use in manual testing.
- **Integrated with CI/CD pipelines** using **Jenkins, Git**.
- **Applied SAFe Agile methodologies** using Jira, Confluence, InVision, fostering collaboration and alignment across teams.

Senior Test Automation Engineer (Contract)

Westpac New Zealand Limited

October 2017 - August 2019

- Led test automation activities, including designing, developing, and maintaining **automation frameworks and scripts**.
 - **Mobile Application Test Automation** using **XCUITest, Appium, Seetest**.
 - **Web Applications** using **Selenium, Cucumber, Cypress**.
 - **API Testing** using **Postman, REST Assured**.
- **Integrated with CI/CD pipelines** using **Jenkins, Git, BitBucket**.
- **Applied SAFe Agile methodologies** using Jira, Confluence, Miro, and InVision, fostering collaboration and alignment across teams.

Manager

Capgemini Australia Pty Limited

June 2015 - October 2017

- **Business & Technology Test Manager** for client ANZ Bank
- Led testing across multiple concurrent **Agile and Waterfall** projects, managing end-to-end test planning, execution, and reporting to ensure alignment with project objectives, quality standards, and deadlines.
- Demonstrated strong **test leadership** by defining and executing robust **testing strategies**, coordinating **cross-functional teams**, and driving quality assurance to achieve successful project outcomes.

- Developed a **web services data extraction and comparison utility**, achieving a **30% reduction** in effort.
- **Managed client relationships** effectively, ensuring timely delivery and **high client satisfaction** for all projects.
- Directed performance appraisals, identified team training needs, set goals, and developed high-performing teams, promoting professional growth and team achievement for a team of 20.
- **Client ANZ:**
- Led and managed end-to-end testing for the implementation of a new card payment scheme (UPI China), ensuring seamless integration with issuer/acquirer systems, payment gateways, and core banking systems.
- Defined the overall test strategy covering functional, integration, regression, certification, and performance testing across multiple workstreams.
- Collaborated with cross-functional teams including business analysts, developers, third-party vendors to validate end-to-end payment flows and transaction lifecycle.
- Oversaw validation of EMV chip, contactless, and e-commerce transaction scenarios, including authorization, clearing, settlement, chargebacks, and dispute processes.
- Managed defect triage, reporting of test metrics, and go-live readiness sign-off with key stakeholders.
- Created and implemented detailed **deployment plans**, performing rigorous **post-implementation verification** to ensure seamless integration and optimal system performance.

Project Manager

Tech Mahindra Ltd

June 2014 - June 2015

- Developed and implemented **mobile automation strategies**, reducing testing time by 50% and increasing test coverage by 40%.
- **Strengthened client relationships**, consistently delivering projects on schedule, exceeding client expectations, and ensuring high levels of customer satisfaction.
- **Led multiple client engagements and projects**, managing resources effectively to achieve on-time, high-quality project delivery.
- **Headed the Mobile Application Test Automation Competency team**, establishing standards for automation excellence.
- Designed and implemented **custom automation solutions** for mobile platforms in response to RFPs/RFIs, aligning with client needs and business goals.
- Conducted **feasibility analyses, cost-benefit assessments, ROI analyses**, and **proof of concepts (POCs)** to facilitate data-driven decision-making and strategic planning.
- **Led recruitment efforts**, identifying and onboarding top talent to build a high-performing team aligned with project and organizational goals.
- Managed team **performance appraisals**, **set strategic goals**, identified **training needs**, and developed **high-performance teams** focused on continuous improvement for a team of 15.

United Health Group, Gurgaon, India
IT Quality Consultant (Project Lead)

January 2010 - June 2014

May 2008 - September 2009

- Led feasibility analysis and **designed automation solutions**, including proof of concepts (POCs), aligning with project objectives.
- **HIPAA EDI files** are standardized formats used in the healthcare industry to electronically exchange **administrative and financial data** between providers, payers, and clearinghouses — in compliance with HIPAA regulations.
- **Saved \$0.5 million** by developing **HIPAA EDI file-to-database** comparison utilities, improving data integrity and ensuring compliance with industry standards.
- Conducted **automation effort estimation, cost-benefit analysis, and ROI assessments**, optimizing resource allocation and improving project efficiency.
- **Managed complex projects in a matrix environment, driving cross-functional collaboration** and ensuring timely delivery of objectives.
- Led **recruitment efforts, performance appraisals, set strategic goals, identified training needs**, and fostered high-performance teams committed to continuous improvement for a team of 20+.
- **Built a three-tier automation framework** to streamline test automation across multiple environments, covering scripting, data management, and execution.
- Led the adoption of automation frameworks across multiple projects by facilitating demos and stakeholder presentations, showcasing functionality and highlighting the benefits for improved testing efficiency and quality.

Sapient Corporation, Gurgaon, India

Senior Associate QA L1

September 2009 - January 2010

June 2005 - April 2008

- Automated report generation, **reducing effort by 75%**, minimizing manual tasks, and enhancing accuracy by lowering error rates.
- Designed and implemented a **QTP-based keyword-driven automation framework** for web applications, increasing test efficiency and accuracy.
- Presented the automation framework to clients, highlighting its capabilities and benefits, and successfully integrated it across multiple projects.
- **Mentored and managed team members**, providing guidance, support, and fostering team development.

Ghaziabad Computer Centre, Ghaziabad, India

Trainee

August 2004 - May 2005

- Designed engaging catalogues to enhance product visibility and customer interest.
 - Assisted in sales by addressing customer inquiries via phone and in-store interactions.
 - Supported the maintenance of computer hardware, ensuring operational efficiency.
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AWARDS & ACHIEVEMENTS

- **Delivered a global webinar** on “**UFT vs Selenium: Mobile Testing Strategy**,” attracting 1,000+ participants.
 - Received **Sustaining Edge Award (2012)** for saving \$0.5 million with automation solutions at United Health Group.
 - Received **Rock Star Award (2009)** for achieving a 75% reduction in reporting generation effort at Sapien Corporation.
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EDUCATION & CERTIFICATIONS

- **Bachelor of Technology in Computer Science & Engineering** from Uttar Pradesh Technical University, 2000-2004.
- **Experitest Seetest Certified Professional**, License 201406231849.